ANSWER KEY & MARKING SCHEME SET 1

Chapter 6: **Perimeter and Area** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 40	Area $= \frac{1}{2} \times$ base $\times$ height.
2	**(a)** 24	Half the product of the legs.
3	**(a)** 32	Find the side first.
4	**(a)** 9	Breadth = area ÷ length.
5	**(d)** 1200	Cost = perimeter $\times$ rate.
6	**(c)** 300	Divide the floor area by the area of one tile.
7	**(b)** 300	1 m = 100 cm.
8	**(b)** 42	Area $= \frac{1}{2} \times$ base $\times$ height.
9	**(a)** 30	Half the product of the legs.
10	**(c)** 44	Find the side first.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Area $= \frac{1}{2} \times b \times h$ (1); $\frac{1}{2} \times 12 \times 9 = 54$ cm^2 (1).	1+1
12	Rectangle area $= 60$ cm^2 (1); Each triangle $= 30$ cm^2 (1).	1+1
13	Perimeter $= 2(30 + 20) = 100$ m (1); Cost $= 100 \times 15 =$ ₹1500 (1).	1+1
14	Area $= \frac{1}{2} \times b \times h$ (1); $\frac{1}{2} \times 16 \times 7 = 56$ cm^2 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Perimeter $= 2(15 + 8) = 46$ cm (1); Area $= 120$ cm^2 (1); Units must be written (1).	1+1+1
16	Legs 6 cm and 8 cm; area $= \frac{1}{2} \times 6 \times 8 = 24$ cm^2 (1); Perimeter $= 6 + 8 + 10 = 24$ cm (1); Units: cm and cm^2 (1).	1+1+1
17	Floor $= 400$ cm by 300 cm (1); Tiles along length 20, along width 15 (1); Number of tiles $= 300$ (1).	1+1+1
	OR: Perimeter $= 2(21 + 10) = 62$ cm (1); Area $= 210$ cm^2 (1); Units must be written (1).	1+1+1
18	Legs 5 cm and 12 cm; area $= \frac{1}{2} \times 5 \times 12 = 30$ cm^2 (1); Perimeter $= 5 + 12 + 13 = 30$ cm (1); Units: cm and cm^2 (1).	1+1+1
	OR: Floor $= 600$ cm by 400 cm (1); Tiles along length 24, along width 16 (1); Number of tiles $= 384$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Area of first part $= 10 \times 6 = 60$ cm^2 (1); Area of second part $= 4 \times 5 = 20$ cm^2 (1); Total area $= 80$ cm^2 (1); Difference $= 40$ cm^2 (1); State the units (1).	1+1+1+1+1
	OR: Outer rectangle: 24 m by 19 m (1); Outer area $= 456$ m^2 (1); Garden area $= 300$ m^2 (1); Path area $= 156$ m^2 (1); State units (1).	1+1+1+1+1
20	Area of first part $= 12 \times 5 = 60$ cm^2 (1); Area of second part $= 6 \times 4 = 24$ cm^2 (1); Total area $= 84$ cm^2 (1); Difference $= 36$ cm^2 (1); State the units (1).	1+1+1+1+1
	OR: Outer rectangle: 36 m by 26 m (1); Outer area $= 936$ m^2 (1); Garden area $= 600$ m^2 (1); Path area $= 336$ m^2 (1); State units (1).	1+1+1+1+1

Mistakes to avoid:

- Using area units (cm^2) for perimeter or linear units (cm) for area.
- Forgetting the $\frac{1}{2}$ in the area of a triangle.
- Using the sloping side instead of the perpendicular height in the triangle area.
- Writing $P = l + b$ for a rectangle (it needs $2(l + b)$) or $A = 2(l + b)$.
- Mixing units: multiplying metres by centimetres without converting first.
- Using area when the question is about fencing (perimeter) and the reverse for paving or tiling.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 6: **Perimeter and Area** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 6	Breadth = area ÷ length.
2	**(d)** 2400	Cost = perimeter × rate.
3	**(b)** 300	Divide the floor area by the area of one tile.
4	**(c)** 700	1 m = 100 cm.
5	**(d)** 63	Area = $\frac{1}{2}\times$ base × height.
6	**(a)** 54	Half the product of the legs.
7	**(d)** 52	Find the side first.
8	**(d)** 12	Breadth = area ÷ length.
9	**(a)** 1800	Cost = perimeter × rate.
10	**(a)** 320	Divide the floor area by the area of one tile.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Rectangle area = 144 cm^2 (1); Each triangle = 72 cm^2 (1).	1+1
12	Perimeter = $2(45+35) = 160$ m (1); Cost = $160 \times 12 =$ ₹1920 (1).	1+1
13	Area = $\frac{1}{2} \times b \times h$ (1); $\frac{1}{2} \times 22 \times 10 = 110$ cm^2 (1).	1+1
14	Rectangle area = 140 cm^2 (1); Each triangle = 70 cm^2 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Perimeter = $2(24+7) = 62$ cm (1); Area = 168 cm^2 (1); Units must be written (1).	1+1+1
16	Legs 9 cm and 12 cm; area = $\frac{1}{2} \times 9 \times 12 = 54$ cm^2 (1); Perimeter = $9+12+15 = 36$ cm (1); Units: cm and cm^2 (1).	1+1+1
17	Floor = 500 cm by 400 cm (1); Tiles along length 10, along width 8 (1); Number of tiles = 80 (1).	1+1+1
	OR: Perimeter = $2(20+12) = 64$ cm (1); Area = 240 cm^2 (1); Units must be written (1).	1+1+1
18	Legs 8 cm and 15 cm; area = $\frac{1}{2} \times 8 \times 15 = 60$ cm^2 (1); Perimeter = $8+15+17 = 40$ cm (1); Units: cm and cm^2 (1).	1+1+1
	OR: Floor = 800 cm by 600 cm (1); Tiles along length 20, along width 15 (1); Number of tiles = 300 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Area of first part = $15 \times 8 = 120$ cm^2 (1); Area of second part = $5 \times 6 = 30$ cm^2 (1); Total area = 150 cm^2 (1); Difference = 90 cm^2 (1); State the units (1).	1+1+1+1+1
	OR: Outer rectangle: 29 m by 22 m (1); Outer area = 638 m^2 (1); Garden area = 450 m^2 (1); Path area = 188 m^2 (1); State units (1).	1+1+1+1+1
20	Area of first part = $14 \times 7 = 98$ cm^2 (1); Area of second part = $8 \times 3 = 24$ cm^2 (1); Total area = 122 cm^2 (1); Difference = 74 cm^2 (1); State the units (1).	1+1+1+1+1
	OR: Outer rectangle: 48 m by 33 m (1); Outer area = 1584 m^2 (1); Garden area = 1000 m^2 (1); Path area = 584 m^2 (1); State units (1).	1+1+1+1+1

Mistakes to avoid:

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- Forgetting the $\frac{1}{2}$ in the area of a triangle.
- Using the sloping side instead of the perpendicular height in the triangle area.
- Writing $P = l + b$ for a rectangle (it needs $2(l+b)$) or $A = 2(l+b)$.
- Mixing units: multiplying metres by centimetres without converting first.
- Using area when the question is about fencing (perimeter) and the reverse for paving or tiling.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 6: **Perimeter and Area** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 1200	1 m = 100 cm.
2	**(d)** 90	Area $= \frac{1}{2} \times$ base $\times$ height.
3	**(d)** 35	Half the product of the legs.
4	**(b)** 68	Find the side first.
5	**(d)** 8	Breadth = area ÷ length.
6	**(a)** 5000	Cost = perimeter $\times$ rate.
7	**(b)** 300	Divide the floor area by the area of one tile.
8	**(b)** 2500	1 m = 100 cm.
9	**(c)** 88	Area $= \frac{1}{2} \times$ base $\times$ height.
10	**(b)** 60	Half the product of the legs.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Perimeter $= 2(60 + 40) = 200$ m (1); Cost $= 200 \times 18 =$ ₹3600 (1).	1+1
12	Area $= \frac{1}{2} \times b \times h$ (1); $\frac{1}{2} \times 30 \times 8 = 120$ cm^2 (1).	1+1
13	Rectangle area $= 144$ cm^2 (1); Each triangle $= 72$ cm^2 (1).	1+1
14	Perimeter $= 2(55 + 25) = 160$ m (1); Cost $= 160 \times 20 =$ ₹3200 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Perimeter $= 2(26 + 9) = 70$ cm (1); Area $= 234$ cm^2 (1); Units must be written (1).	1+1+1
16	Legs 12 cm and 16 cm; area $= \frac{1}{2} \times 12 \times 16 = 96$ cm^2 (1); Perimeter $= 12 + 16 + 20 = 48$ cm (1); Units: cm and cm^2 (1).	1+1+1
17	Floor $= 300$ cm by 200 cm (1); Tiles along length 30, along width 20 (1); Number of tiles $= 600$ (1).	1+1+1
	OR: Perimeter $= 2(28 + 15) = 86$ cm (1); Area $= 420$ cm^2 (1); Units must be written (1).	1+1+1
18	Legs 15 cm and 20 cm; area $= \frac{1}{2} \times 15 \times 20 = 150$ cm^2 (1); Perimeter $= 15 + 20 + 25 = 60$ cm (1); Units: cm and cm^2 (1).	1+1+1
	OR: Floor $= 1000$ cm by 500 cm (1); Tiles along length 20, along width 10 (1); Number of tiles $= 200$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Area of first part $= 20 \times 9 = 180$ cm^2 (1); Area of second part $= 7 \times 5 = 35$ cm^2 (1); Total area $= 215$ cm^2 (1); Difference $= 145$ cm^2 (1); State the units (1).	1+1+1+1+1
	OR: Outer rectangle: 41 m by 28 m (1); Outer area $= 1148$ m^2 (1); Garden area $= 770$ m^2 (1); Path area $= 378$ m^2 (1); State units (1).	1+1+1+1+1
20	Area of first part $= 18 \times 10 = 180$ cm^2 (1); Area of second part $= 9 \times 4 = 36$ cm^2 (1); Total area $= 216$ cm^2 (1); Difference $= 144$ cm^2 (1); State the units (1).	1+1+1+1+1
	OR: Outer rectangle: 60 m by 40 m (1); Outer area $= 2400$ m^2 (1); Garden area $= 1500$ m^2 (1); Path area $= 900$ m^2 (1); State units (1).	1+1+1+1+1

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- Mixing units: multiplying metres by centimetres without converting first.

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